

Table 3: The Big 5

Details	AWS IoT	Microsoft Azure IoT	Google Cloud IoT	IBM Watson IoT	Oracle IoT
Security	Link Encryption (TLS), Authentication (Sig V4, X.509)	Link Encryption (SSL/TSL)	SSL/TLS	Link Encryption (TLS), Authentication (IBM cloud SSO), Identity Management (LDAP)	REST API
Data analytics	Real-Time analytics (Rule engine, Kinesis, AWS Lambda)	Real-Time analytics	Real-Time analytics (Cloud IoT Core)	Real-Time analytics (IBM IoT Real-Time insights)	Real-Time analytics
Data format	JSON	JSON	JSON	JSON, CSV	CSV, REST API
Application Environment	Java, C, NodeJs, Javascript, Python, SDK for Arduino, iOS, Android	.Net, UWP, Java, C, NodeJS, Ruby, Android, iOS	Go, Java, Python, .NET, NodeJS	C, Python, Java, NodeJS	Java, iOS, Javascript, C, Android
Protocols	MQTT, HTTP1.1	MQTT, HTTP, AMQP	MQTT	MQTT, HTTPS	MQTT, HTTP
Visualisation	AWS IoT dashboard	Web Dashboard	Google data studio (Dashboard)	Web Dashboard	Web Dashboard
Edge Computing Options	AWS Greengrass	IoT Edge	Google Edge TPU	IBM Edge Application Manager	Oracle Roving Edge
Pricing	AWS allows you to pay for the services you use. You can calculate your pricing using the AWS Pricing Calculator.	A free version, basic version, and a standard version are available. Pricing varies for each.	Cloud IoT Core is priced according to the data volume used in a calendar month.	Three plans are available, depending on the service selected: Connection, Analytics, or Blockchain.	Oracle offers uniform pricing across all global regions and also provides a cost estimator similar to AWS.
Website	https://aws.amazon.com/iot-core/?nc=sn&loc=0	https://azure.microsoft.com/en-in/services/iot-hub	https://cloud.google.com/solutions/iot	https://internetofthings.ibmcloud.com	https://www.oracle.com/in/internet-of-things

12. Edge support

IoT is moving towards edge computing, AI, and ML. Hence, the support for such features is important, even if your solution does not employ them at this point in time. There are many platforms in the market that are dedicated to analytics and edge intelligence, but even if your selected platform is not one of them, it is a good sign if it still accommodates new topologies and utilises edge intelligence.

13. Bandwidth and range

High bandwidth means low latency. For fast movement of data and good connectivity, the system should consume extremely little power, have a huge range, and be able to transmit a large volume of data. Hence, bandwidth management for IoT devices should not be overlooked. At the same time, the system should have a good enough range to accommodate your hardware—there will always be a tradeoff.

14. Time-to-market and platform migration

To reduce time-to-market, some IoT platform providers offer quick-start packages for new customers, and some offer drag-and-drop functionality to generate simple applications. Overall, the IoT platform you select must allow you to go to market in your projected timeline. Furthermore, in the unlikely event that you need to change your platform after going to market, the transition needs to be smooth, and your initial platform needs to permit this migration in a hassle-free manner.

15. Pricing model

This is one of the most important considerations. Most IoT platforms provide multiple pricing options depending on features, similar to how

Gmail provides services to enterprises. But some might try to sneak in some additional costs that you might overlook while selecting! Some IoT platforms might charge according to the number of reads and writes; so it becomes costlier as your company grows! Also, you can try to check if the company is flexible enough to accommodate your needs and modify the pricing accordingly.

Comparison of some IoT platforms

Let us take a look at a few IoT platforms and compare their features and pricing as given in Table 1 to 3.

Once all relevant platforms are analysed according to the fifteen points mentioned above, you will be able to decide which one suits your application the best.  **END**

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